Quarl Table 2: NAM total gates with native RL-ZX results in the last column

Circuit	Orig.	Nam	VOQC	Qiskit	Tket	Quartz +RM	Quartz no RM	Quarl +RM	Quarl no RM	RL-ZX previous	RL-ZX native new / pending
tof_3	45	35	35	43	39	35	35	**33**	**33**	**33**	**33**
barenco_tof_3	58	40	40	54	49	40	46	**35**	36	37	38
mod5_4	63	51	51	61	57	37	39	24	24	**22**	23
tof_4	75	55	55	71	64	55	55	**51**	**51**	55	53
barenco_tof_4	114	72	72	106	97	72	90	**62**	**62**	69	64
tof_5	105	75	75	99	89	75	75	**69**	**69**	75	71
mod_mult_55	119	91	92	115	103	90	94	**84**	**84**	88	88
vbe_adder_3	150	89	89	189	130	89	112	**71**	**71**	76	74
barenco_tof_5	170	104	104	158	145	104	134	**90**	96	101	99
csla_mux_3	170	155	158	249	149	148	146	**141**	**141**	150	148
rc_adder_6	200	140	152	226	172	152	170	**134**	148	157	156
gf2^4_mult	225	187	186	212	205	176	215	**160**	163	168	167
tof_10	255	175	175	239	214	175	175	**169**	**169**	175	173
mod_red_21	278	180	184	256	223	198	268	179	**177**	195	188
hwb6	259	-	209	258	217	214	250	**193**	196	204	198
gf2^5_mult	347	296	287	327	319	274	334	**257**	266	272	268
csum_mux_9	420	266	280	420	365	256	420	**224**	282	299	289
barenco_tof_10	450	264	264	418	385	264	450	**236**	274	257	254
qcla_com_7	443	284	269	498	377	288	431	**258**	277	273	259
ham15-low	443	-	348	421	392	360	435	323	**322**	384	362
gf2^6_mult	495	403	401	464	453	391	485	**352**	386	378	375
qcla_adder_10	521	399	416	630	444	404	518	**380**	391	432	419
gf2^7_mult	669	555	543	627	614	531	657	**484**	530	514	513
gf2^8_mult	883	712	706	819	806	703	883	**648**	755	667	665
qcla_mod_7	884	**624**	678	933	759	652	884	629	664	694	677
adder_8	900	606	596	1001	802	624	874	560	669	568	**541**
Geo. mean reduction	-	28.0%	27.0%	-0.6%	13.1%	28.3%	12.9%	35.8%	32.6%	30.8%	32.2%

Lower is better. Blue = new native-only cohort; independent replay audit still in progress.
Previous paper / RL-ZX columns preserved. Seed 11; each target stops after 3 h without a strict gate improvement.
Preprocessing, extraction, hardware and training budgets differ; counts do not isolate learning benefit.
Source: arxiv.org/html/2307.10120v1#S6.T2. RM = rotation merging. Bold = row minimum across optimizers.